WEEK 3 · LESSON 14 OF 24 · HANDOUT

Matrices: Many Calculations at Once

Study time: **about 50 minutes** (reading, video and activity)

▶ **Watch:** https://www.youtube.com/watch?v=kYB8IZa5AuE — open the lesson online to watch inside the course.

Your goal in this lesson: Describe a matrix by its shape, multiply a matrix by a vector and by another matrix, use the transpose, and see how one matrix multiplication makes predictions for a whole dataset.

A matrix is a table of numbers

A **matrix** is a grid of numbers with rows and columns. A dataset is a matrix: one **row per example**, one **column per feature**.

```
import numpy as np

# columns: size_m2, bedrooms, age_years
X = np.array([
    [120, 3, 10],
    [80, 2, 25],
    [200, 4, 3],
    [95, 2, 15],
])
print(X.shape)   # (4, 3): 4 houses, 3 features
```

We usually write the shape as **rows × columns**: X is a 4 × 3 matrix.

Matrix × vector = predictions for every row

Suppose a simple model prices a house (in thousands of RWF) as:

> *price = 250 × size + 4,000 × bedrooms − 300 × age*

Put the weights in a vector and multiply:

```
w = np.array([250, 4000, -300])
prices = X @ w
print(prices)     # one prediction per house
```

Each row of X is dot-multiplied with w. **One line predicts the whole dataset.** This is exactly what a linear model — and each layer of a neural network — does.

The shape rule

You can multiply A @ B only when **the number of columns of A equals the number of rows of B**:

> *(4 × 3) @ (3,) → (4,) ✓ (4 × 3) @ (3 × 2) → (4 × 2) ✓ (4 × 3) @ (4 × 3) → error ✗*

When you get a shape error in AI code, write the shapes down — the fix is almost always here.

Matrix × matrix

Two different models (two sets of weights) at once:

```
W = np.array([
    [250, 200],
    [4000, 6000],
    [-300, -100],
])          # 3 × 2: one column per model
print(X @ W)  # 4 × 2: every house priced by both models
```

A neural-network layer with 3 inputs and 2 neurons is exactly this: `outputs = X @ W + b`.

The transpose

The **transpose** swaps rows and columns: a 4 × 3 matrix becomes 3 × 4.

```
print(X.T.shape)    # (3, 4)
print(X.T @ X)      # a 3 × 3 matrix used in linear regression formulas
```

Matrices transform space

3Blue1Brown's video shows the deeper idea: a 2 × 2 matrix **moves every point** of the plane — rotating, stretching or flipping it.

```
rotate_90 = np.array([[0, -1],
                      [1,  0]])
stretch = np.array([[2, 0],
                    [0, 1]])
point = np.array([1, 2])
print(rotate_90 @ point)   # [-2  1]
print(stretch @ point)     # [2 2]
```

Neural networks learn matrices that transform data step by step until the answer becomes easy to read off.

The identity matrix

The identity matrix `I` has 1s on the diagonal and 0s elsewhere. Multiplying by it changes nothing — like multiplying a number by 1.

```
I = np.eye(3)
print(np.allclose(X @ I, X))   # True
```

Key terms

- **Matrix** — a 2-D grid of numbers (rows × columns).
- **Shape** — rows × columns, e.g. 4 × 3.
- **Matrix multiplication** (`@`) — dot products of rows with columns.
- **Shape rule** — columns of A must equal rows of B.
- **Transpose** (`.T`) — swap rows and columns.
- **Identity matrix** — the matrix that leaves vectors unchanged.

Remember this: A dataset is a matrix, a model's weights are a vector or matrix, and `X @ w` makes predictions for every example at once. Check shapes whenever multiplication fails.

Name: Date:

Activity

Task 1

Add a fifth house to X and predict its price with `X @ w`.

Task 2

Change the weights so that age does not matter (weight 0). Which house changes most?

Task 3

Predict the shape of `X @ W` when X is 100 × 5 and W is 5 × 8. Check it with `np.ones` arrays.

Task 4

Use the stretch and rotation matrices one after the other on the point [1, 2]. Does the order matter?

Self-check

Think about your answers here, then take the quiz online — it is marked instantly and explains every answer.

1. A dataset has 500 examples and 6 features. What is its shape?
   A. 500 × 500
   B. 500 × 6
   C. 6 × 500
   D. 3000 × 1
2. What does `X @ w` compute for a data matrix X and weight vector w?
   A. A prediction (weighted sum) for every row at once
   B. The sum of all numbers
   C. The average row
   D. The transpose
3. Can you multiply a (4 × 3) matrix by a (3 × 2) matrix?
   A. Yes, the result is 3 × 3
   B. Yes, the result is 4 × 2
   C. Yes, the result is 4 × 3
   D. No
4. What does the transpose do?
   A. Doubles every number
   B. Sorts the matrix
   C. Swaps rows and columns
   D. Deletes rows
5. What is special about the identity matrix?
   A. Multiplying by it leaves a vector unchanged
   B. It cannot be multiplied
   C. It is all zeros
   D. It rotates by 90°
6. What does one layer of a neural network compute?
   A. outputs = X @ W + b (then an activation)
   B. A pie chart
   C. Only an average
   D. A sorted list

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